Activity: Titan2D on VICTOR

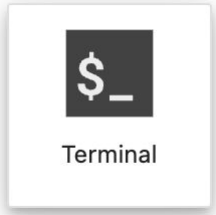
TITAN2D is a geoflow simulation software application, specifically used for granular flows. It is designed for dry granular avalanches, and uses a depth-averaged model of the conservation laws and an adaptive grid Godunov solver.

References:

Patra, A. K., Bauer, A. C., Nichita, C. C., Pitman, E. B., Sheridan, M. F., Bursik, M., et al. (2005). Parallel adaptive numerical simulation of dry avalanches over natural terrain. *J. Volcanol. Geotherm. Res*. 139, 1–21. doi: 10.1016/j.jvolgeores.2004.06.014

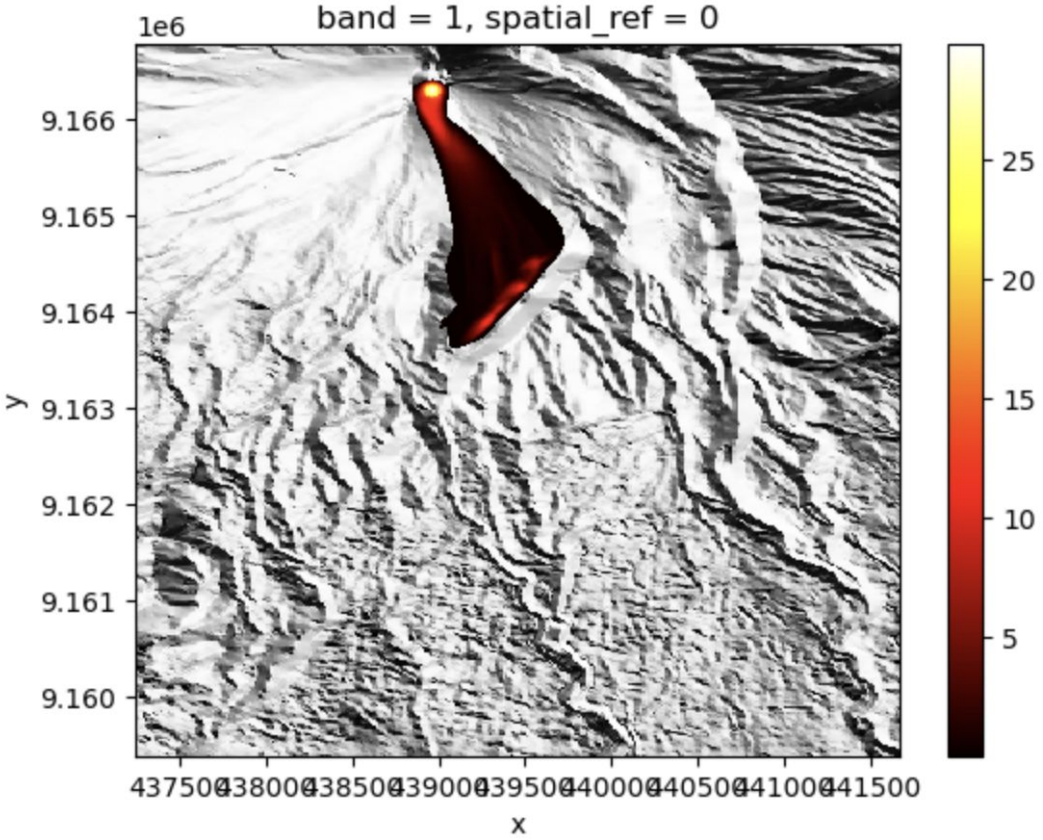
Patra, A., Bevilacqua, A., Akhavan-Safaei, A., Pitman, E. B., Bursik, M., & Hyman, D. (2020). Comparative analysis of the structures and outcomes of geophysical flow models and modeling assumptions using uncertainty quantification. Frontiers in Earth Science, 8. https://doi.org/10.3389/feart.2020.00275

Activity: Titan2D on VICTOR



- Login to VICTOR and select a small server size.
- In The Launcher, open the terminal using the icon:
- Type *victor setup* and press enter.
- Select TITAN2D from the list that appears by typing its corresponding number (19) and pressing enter.
- In the home directory (on the left of your screen) the Titan2D folder will load. Double click on the folder to open it.
- Within the Titan2D folder, you will see the Notebook titan2d.ipynb open the Titan2D Notebook by double clicking on it.
- Run the model:
- Save your visual outputs by holding shift and right clicking on the graph, then selecting “Save Image As…”

Example Output:



Notes on Running the Titan2D Model:

[7]:

```
if a cell finishes running and you see a red
message beneath it:
```

review the message within the red, if it does not read ‘error’ within the first few lines, it is considered a warning, and the notebook can continue running!

If you receive an unexpected error message:
Consult the VICTOR forum
or
Contact the VICTOR team at victor@ldeo.columbia.edu - include screenshots in your email!